II Measure of Dispersion Suggestions:

- What is dispersion? What are the absolute and relative measures of dispersion? (RU-ICE: 2013)
- Define standard deviation. Also mention its merits and demerits.
- Find mean and variance for first n natural number. (RU-ICE: 2013; CSE -2017)
- Calculate mean and standard deviation using the given data

Mid Value	Frequency
10	3
15	14
20	29
25	35
30	41
35	20
40	11
45	2

(RU-ICE: 2013)

- What do you mean by absolute and relative measure of dispersion? (RU-ICE: 2013)
- Show that the standard deviation for first n natural number is

$$\delta^2 = \sqrt{\frac{n^2-1}{12}}$$

(RUET-civil: 2014)

- If X and S denote the mean and standard deviation of $x_1, x_2, \ldots\ldots, x_n$ then show that $X\sqrt{(n-1)} \geq S$ (RU-ICE: 2014) ; CSE -2017
- For two observations, standard deviation is half of the range , Prove. (RU-ICE: 2011)
- Which measure of dispersion do you consider to be the best and why? (RU-ICE: 2014)
- What is coefficient of variation? When do we use it ? (RU-ICE: 2014)
- Given below the monthly income s in Tk for 10 families
  10450, 17416,6517,13600,14821,9226,152936,11800,18500,15225
  Use the above data to compute mean deviation from mean and mean deviation from median. (RU-ICE: 2012)

- Calculate mean standard deviation and coefficient of variation using the given data

Class interval	Frequency
0-5	2
5-10	5
10-15	7
15-20	13
20-25	21
25-30	16
30-35	8
35-40	3

(RU-ICE: 2011)

- Series of two golfers for 12 rounds were as follows:

Golfer A	74	75	78	81	84	73	68	71	76	80	67	72
Golfer B	91	84	81	88	86	89	79	81	83	78	80	82

Find which golfer may be considered to be a more consistent players? (RUET: 2013-civil)

- The following are the scores of two batsman A and B in a series of innigs

Player A	45	101	8	80	10	120	35	15
Player B	48	10	35	22	50	60	25	34

Find who is the better scorer and who is more consistent players? (RUET: 2012-civil)

- Define skewness and kurtosis? What are their types and measures? Calulalate coefficient of skewness and Kurtosis for the following data

Values	12	24	36	78	60	72	84
Frequency	8	14	18	36	30	20	10

(RUET: 2014-civil)

- Define moments. Distinguish between raw moments and central moments. Mention the relationship between raw moments and central moments for the first four moments.

- The first four moments of a distribution about the value 5 of the variable are 2,20,40,50 . Find the moment about mean .Mean and variance. (RUET: 2012-civil)